Cont. shear stress due to Torsion

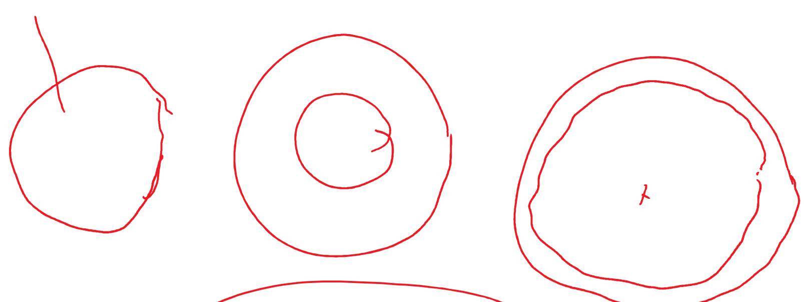
$\phi = \theta$: Angle of deformation

$$\boxed{\frac{\theta G}{l} = \frac{T}{J} = \frac{\tau}{r}}$$

C

$$\tau = \frac{T}{J} r$$

$$J = I_x + I_y$$



$$J = 2I$$

Torque

length

$\theta = Tl$

$$\boxed{\theta = \phi = \frac{T\ell}{JG}}$$

Modulus of Rigidity

$$G = \frac{E}{2(1+\nu)}$$

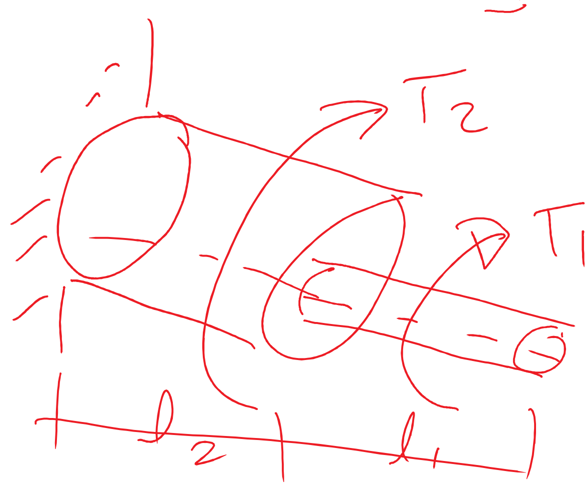
$E_{st} = (200-210)\,Gpa$

$E_{C.I} = (100-120)\,Gpa$

$G_{st} \approx 80\ Gpa$

ν for st $\rightarrow$ 0,3

$$\theta = \phi = \sum_{i=1}^{n} \frac{T_i \ell_i}{J_i G_i}$$



kW

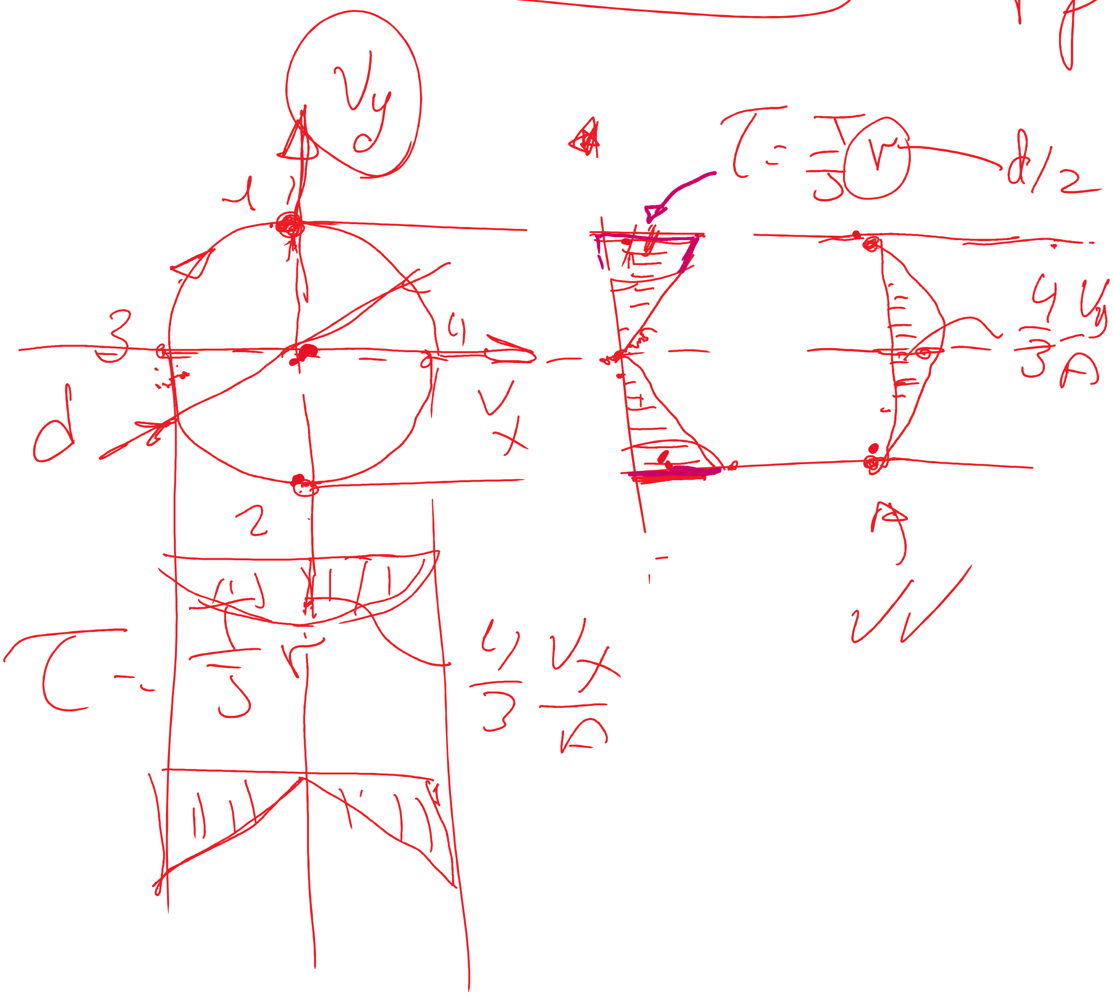
x_2 | x_1

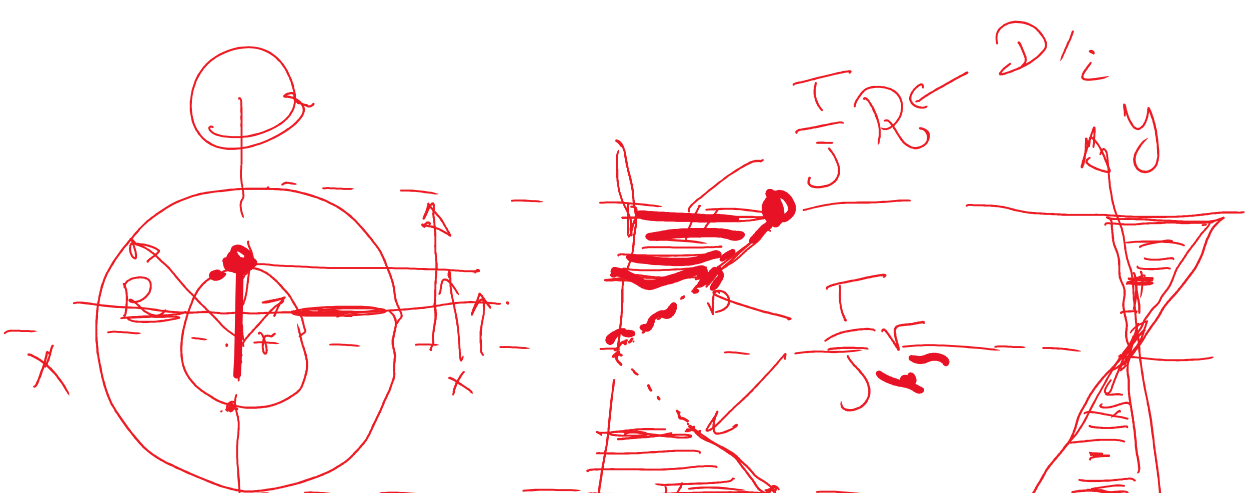
$$\boxed{T = 9550 \frac{\text{Power}}{N}}$$

N.m.

kW

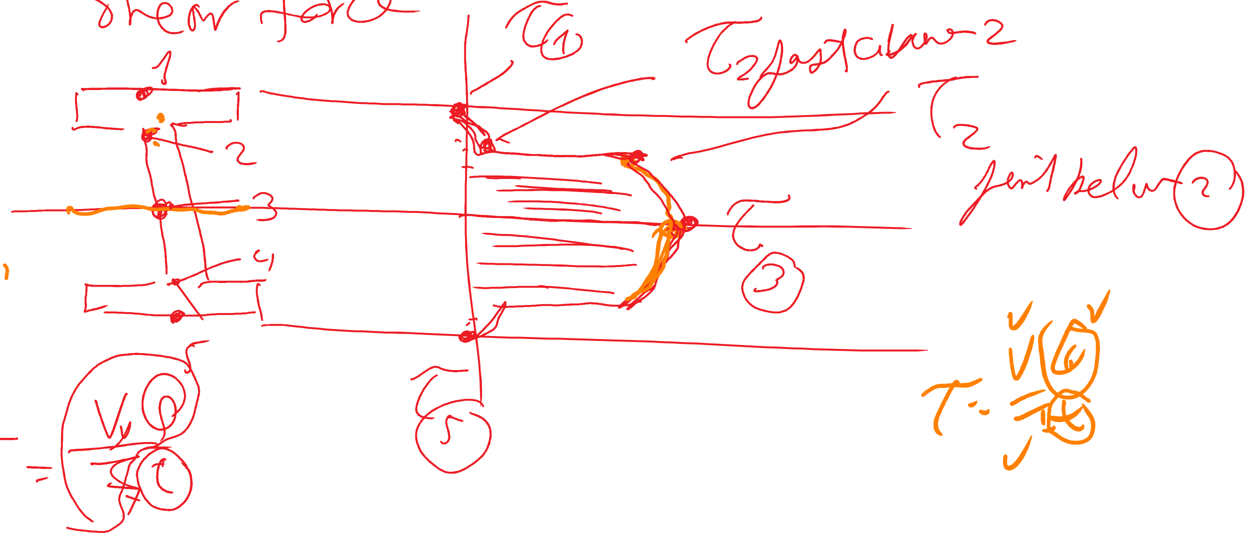
rpm





$$\sigma = \pm \frac{M_x}{I_x} y$$

Cont. ⇒ Shear stress due to shear force



$$\tau = \frac{V_y Q}{I t}$$

$$\tau = \frac{V Q}{I t}$$